

CSA 1718 – Artificial Intelligence

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Prolog Program for Medical Diagnosis

Aim

To write and implement a simple Prolog program for medical diagnosis based on given symptoms.

Objective

- To understand rule-based diagnosis.
- To represent symptoms as facts.
- To use logical rules to determine a diagnosis.

Theory

A medical diagnosis system can be represented using facts and rules. Symptoms are represented as facts, and a disease is inferred when the required symptoms satisfy a rule.

This is a simplified academic example and is not intended for actual medical diagnosis.

Algorithm

1. Define the symptoms as facts.
2. Define rules connecting symptoms to diseases.
3. Give a diagnosis as a query.
4. Prolog checks the required conditions.
5. If all conditions are satisfied, it returns true.

Program

symptom(fever).

symptom(cough).

symptom(body_pain).

diagnosis(flu) :-

symptom(fever),

```
symptom(cough),  
symptom(body_pain).
```

```
diagnosis(cold) :-  
    symptom(cough).
```

Query

```
diagnosis(flu).
```

Output

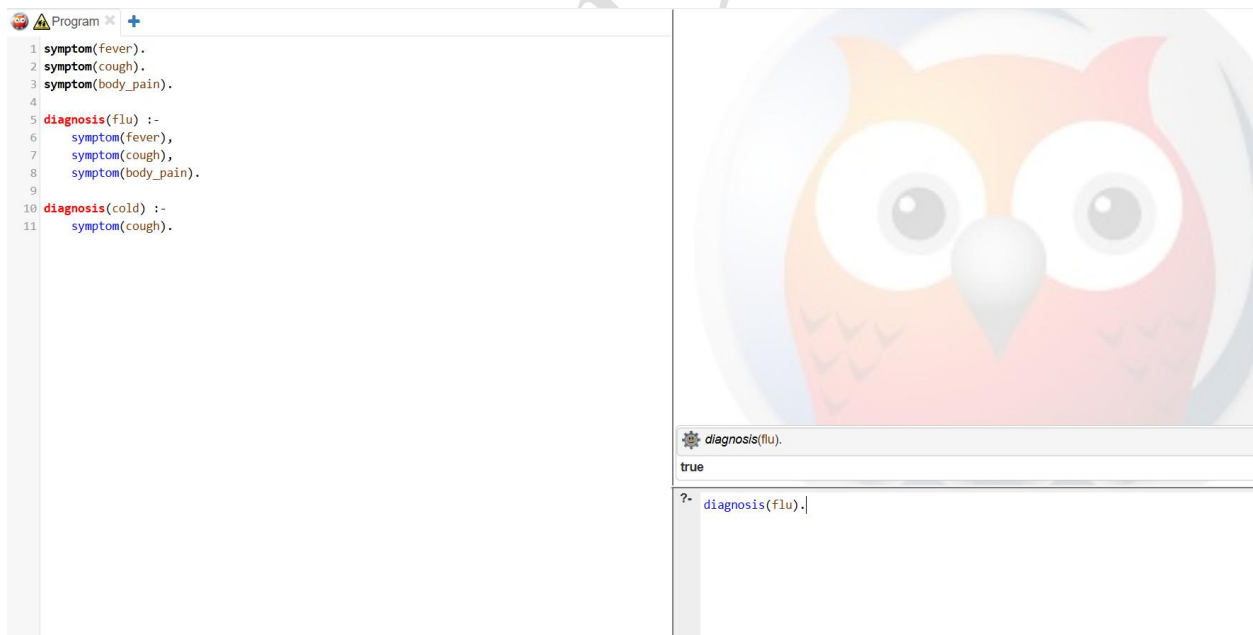
true.

Another query:

```
diagnosis(cold).
```

Output:

true.



Result

Thus, a simple rule-based medical diagnosis program was successfully implemented using Prolog.